M3 Modelling with probability A level only

Name: ____________________

Class: ____________________

Date: ____________________

Time: **39 minutes**

Marks: **33 marks**

Comments:

Q1.

In the South West region of England, 100 households were randomly selected and, for each household, the weekly expenditure, £X, per person on food and drink was recorded.

The maximum amount recorded was £40.48 and the minimum amount recorded was £22.00
The results are summarised below, where x̄ denotes the sample mean.

$$\sum x = 3046.14 \qquad \sum (x - \bar{x})^2 = 1746.29$$

(a) (i) Find the mean of X
Find the standard deviation of X

(2)

(ii) Using your results from part **(a)(i)** and other information given, explain why the normal distribution can be used to model X.

(2)

(iii) Find the probability that a household in the South West spends less than £25.00 on food and drink per person per week.

(1)

(b) For households in the North West of England, the weekly expenditure, £Y, per person on food and drink can be modelled by a normal distribution with mean £29.55

It is known that P(Y < 30) = 0.55

Find the standard deviation of Y, giving your answer to one decimal place.

(3)
(Total 8 marks)

Q2.

The length, L centimetres, of *Slimline* bin liners may be modelled by a normal distribution with a mean of 69.5 and a standard deviation of 0.55.

(a) Determine:

(i) P(L < 70);

(3)

(ii) P(69 < L < 70);

(3)

(iii) P(L = 70).

(1)

(b) Determine the maximum length exceeded by 90% of bin liners.

(4)

(c) The bin liners are sold in packets of 20, and those in each packet may be considered to be a random sample.

Determine the probability that the mean length of the bin liners in a packet is greater than 69.25 cm.

(4)
(Total 15 marks)

Q3.

Chopped lettuce is sold in bags nominally containing 100 grams.

The weight, X grams, of chopped lettuce, delivered by the machine filling the bags, may be assumed to be normally distributed with mean μ and standard deviation 4.

(a) Assuming that μ = 106, determine the probability that a randomly selected bag of chopped lettuce:

(i) weighs less than 110 grams;

(3)

(ii) is underweight.

(3)

(b) Determine the minimum value of μ so that at most 2 per cent of bags of chopped lettuce are underweight. Give your answer to one decimal place.

(4)
(Total 10 marks)